Fundamentals Chapter 12

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1. Fix $\varepsilon > 0$. We know that there exists N_1, N_2 such that if $n > N_1$ then $|x_n| < \varepsilon/2$ and if $n > N_2$ then $|y_n| < \varepsilon/2$. Let $n > \max(N_1, N_2)$. Then

$$|x_n + y_n| \leq |x_n| + |y_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Furthermore, we know there exists N_1, N_2 such that if $n > N_1$ then $|x_n| < \sqrt{\varepsilon}$ and if $n > N_2$ then $|y_n| < \sqrt{\varepsilon}$. So if $n > \max(N_1, N_2)$, then

$$|x_n y_n| = |x_n||y_n| < \sqrt{\varepsilon}\sqrt{\varepsilon} = \varepsilon.$$

So $x_n + y_n$ and $x_n y_n$ converge to 0.

2a. Since x_n is Cauchy, it is bounded, and let $M > |x_n|$ for all n. Now, fix $\varepsilon > 0$. We know that there exists N such that $|y_n| < \varepsilon/M$, so if $n > N$ then

$$|x_n y_n| = |x_n||y_n| < |x_n|\frac{\varepsilon}{M} < M\frac{\varepsilon}{M} = \varepsilon.$$

2b. Assume $x_n \to 0$. Then, given $\varepsilon > 0$, there exists N such that if $n > N$ then $|x_n| < \varepsilon$. Since $|x_n| = ||x_n||$, $||x_n|| < \varepsilon$ as well, so $|x_n| \to 0$.

Now, assume $|x_n| \to 0$. Then given $\varepsilon > 0$, there exists N such that if $n > N$ then $||x_n|| < \varepsilon$, which means $|x_n| < \varepsilon$ as well.

3. Since x_n is Cauchy, given $\varepsilon > 0$, there exists N such that if $n, m > N$ then $|x_n - x_m| < \varepsilon$. Let $m = k_n$, so that $n, k_n > N$ (since $k_n \geq n$ by definition of a subsequence). Then we have that $|x_n - x_{k_n}| < \varepsilon$, so $(x_n - x_{k_n}) \to 0$.

4a. To prove this, we show that if $x_n \not\to x$ then there exists a subsequence that has no subsequences converging to x.

If $x_n \not\to x$, then there exists ε such that, for all N, there exists $n > N$ such that $|x_n - x| \geq \varepsilon$. To construct a subsequence x_{k_n} with no subsequences converging to x, find the first n where $|x_n - x| \geq \varepsilon$, and let $k_1 = n$. To continue the process infinitely, let $N = k_i$, and find the next $n > N$ so that $|x - x_n| \geq \varepsilon$, and let $k_{i+1} = n$. This can be continued infinitely because $x_n \not\to x$.

Since there exists some ε such that every element of this subsequence is at least ε away from x, it has no subsequences that converge to x.

4b. Let $x_n = (-1)^n$. Any subsequence of x_n has infinitely many 1s or -1s (or both). So we can make a further subsequence by choosing only 1s or only -1s, and it will converge.

5a. We need to show that if $y > x$ then $\frac{y}{1+y} - \frac{x}{1+x} > 0$. Simplifying this, we get

$$\frac{y}{1+y} - \frac{x}{1+x} = \frac{y(1+x) - x(1+y)}{(1+y)(1+x)} = \frac{y-x}{(1+y)(1+x)} > 0.$$

So $f(x) = \frac{x}{1+x}$ is increasing.

5b. If $x = y$ then $d^*(x,y) = \frac{0}{1+0} = 0$. If $d^*(x,y) = 0$, then $\frac{d(x,y)}{1+d(x,y)} = 0$ so $d(x,y) = 0$ and $x = y$.

Now, we prove the triangle inequality. Let $a = d(x,y), b = d(x,z), c = d(y,z)$. Given $a \leq b + c$, we wish to show that $\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c}$. Simplifying this, we get

$$\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c},$$
$$a(1+b)(1+c) \leq b(1+a)(1+c) + c(1+a)(1+b),$$
$$abc + ab + ac + a \leq abc + ab + bc + b + abc + ac + bc + c,$$
$$a \leq abc + 2bc + b + c.$$

Since $a \leq b + c$, we know this is true, so the original inequality is true as well.

5c. Assume that x_n is Cauchy with respect to d. Then, given $\varepsilon > 0$, there exists N such that if $n, m > N$ then $d(x_n, x_m) < \varepsilon$. We know that $\frac{d(x_n,x_m)}{1+d(x_n,x_m)} < d(x_n, x_m) < \varepsilon$, so x_n is Cauchy with respect to d^* as well.

Now, assume x_n is Cauchy with respect to d^*. Fix $\varepsilon > 0$. Then there exists N such that if $n, m > N$ then $d^*(x_n, x_m) < \frac{\varepsilon}{1+\varepsilon}$. Then we have that

$$d^*(x_n, x_m) = \frac{d(x_n, x_m)}{1 + d(x_n, x_m)} < \frac{\varepsilon}{1+\varepsilon},$$
$$d(x_n, x_m)(1+\varepsilon) < \varepsilon(1 + d(x_n, x_m)),$$
$$d(x_n, x_m) + \varepsilon d(x_n, x_m) < \varepsilon + \varepsilon d(x_n, x_m),$$
$$d(x_n, x_m) < \varepsilon.$$

So x_n is Cauchy with respect to d^*.

6a. First we show that $x = y$ iff $(d_1 + d_2)(x, y) = 0$. If $x = y$ then $d_1(x, y) + d_2(x, y) = 0 + 0 = 0$. If $(d_1 + d_2)(x, y) = 0$, then since $d_1, d_2 \geq 0$, we have that $d_1(x, y) = d_2(x, y) = 0$ so $x = y$.

Now, we show the triangle inequality. We know that

$$d_1(x, y) \leq d_1(x, z) + d_1(y, z),$$

$$d_2(x, y) \leq d_2(x, z) + d_2(y, z).$$

Adding the two inequalities gives us

$$(d_1 + d_2)(x, y) \leq (d_1 + d_2)(x, z) + (d_1 + d_2)(y, z),$$

as desired.

6b. First, we show that $x = y$ iff $\left|\frac{1}{x} - \frac{1}{y}\right| = 0$. If $x = y$ then $|1/x - 1/y| = 0$. If $|1/x - 1/y| = 0$ then $\frac{1}{x} = \frac{1}{y}$, so $x = y$.

Now, we show that

$$\left|\frac{1}{x} - \frac{1}{y}\right| \leq \left|\frac{1}{x} - \frac{1}{z}\right| + \left|\frac{1}{y} - \frac{1}{z}\right|.$$

If we let $a = \frac{1}{x} - \frac{1}{z}$ and $b = \frac{1}{z} - \frac{1}{y}$, the inequality simplifies to $|a + b| \leq |a| + |b|$, which we know is true, so the original inequality holds.

8a. Assume $|x| < \varepsilon$. Consider the case where $x \geq 0$. Then $|x| = x < \varepsilon$, so $0 \leq x < \varepsilon$ and the condition $-\varepsilon < x < \varepsilon$ is satisfied. Now, consider the case where $x < 0$. Then $|x| = -x < \varepsilon$, so $x > -\varepsilon$ and $-\varepsilon < x < 0$, so in either case $-\varepsilon < x < \varepsilon$ is satisfied.

Now, assume $-\varepsilon < x < \varepsilon$. If $x \geq 0$, $|x| = x < \varepsilon$ so $0 \leq |x| < \varepsilon$. If $x < 0$, $|x| = -x$, and since $x > -\varepsilon$, we know that $-x < \varepsilon$, so $|x| < \varepsilon$. So in both cases, $|x| < \varepsilon$ is satisfied.

8b. Squaring both sides gives us

$$(|x| - |y|)^2 \leq (x - y)^2,$$

$$x^2 + y^2 - 2|xy| \leq x^2 + y^2 - 2xy,$$

$$xy \leq |xy|,$$

which is true. Equality holds when $xy = |xy|$, or x and y have the same sign.

8c. Given $\varepsilon > 0$, there exists N such that if $n > N$, $|x_n - x| < \varepsilon$. By the reverse triangle inequality, $||x_n| - |x|| \leq |x_n - x| < \varepsilon$, so $|x_n| \to |x|$.

The converse does not hold. For example, take $x_n = (-1)^n$. We have that $|x_n| = 1$ converges to 1 but x_n does not converge.

10. For contradiction, assume $x > y$ and let $\varepsilon = x - y$. We know that there exists N such that if $n > N$ then $|x_n - x| < \varepsilon = x - y$. If $x_n \geq x$ then $x_n > y$, which is a contradiction, so $x_n < x$. That means $|x_n - x| = x - x_n$, so we have that $x - x_n < x - y$, or $y < x_n$, which is a contradiction. So $x \leq y$.

To show that it is possible that $x = y$, let $x_n = 1 - \frac{1}{2^n}$, so that $x = 1$. We also have that $x_n < 1$ for all n, so we can let $y = 1$, so $x = y$.

11. Since x_n doesn't converge to 0, there exists $\varepsilon > 0$ such that for all N, there is some $n \geq N$ with $|x_n| \geq \varepsilon$. This means there exists a subsequence x_{m_k} such that $|x_{m_k}| \geq \varepsilon$ for all k.

Now, assume x_n has infinitely many zeroes, so that there is an increasing sequence, n_k, such that $x_{n_k} = 0$ for all k. Because x_n is Cauchy, there exists N such that for all $n, m \geq N$, $|x_m - x_n| < \varepsilon/2$. From the subsequence x_{m_k}, choose $m_k > N$, and from the subsequence x_{n_k}, choose $n_k > N$. Then $|x_{n_k} - x_{m_k}| = |x_{m_k}| \geq \varepsilon$. But from earlier, it must be $< \varepsilon/2$. This is a contradiction, so x_n cannot have infinitely many 0s.

12. Let $x_n = e^{-n}$ and $y_n = \frac{\sin n}{n}$. They are both Cauchy, but $x_n^{y_n} = e^{-\sin n}$ isn't.